DRUGS · ANTIFUNGALS AND ANTIVIRALS

Cidofovir

Cidofovir is a nucleotide analogue antiviral injected into papillomas at endoscopic debulking as an adjuvant treatment for aggressive recurrent respiratory papillomatosis. Its use has fallen with growing experience of systemic bevacizumab and HPV vaccination, and with concerns about dysplasia and systemic toxicity.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

CIDOFOVIR

	SUMMARY
Class	Acyclic nucleotide analogue of cytosine
How it works	Inhibits viral DNA polymerase after cellular phosphorylation (no viral kinase needed); in HPV-infected tissue, induces apoptosis
ENT uses	Adjuvant intralesional treatment of recurrent respiratory papillomatosis needing frequent surgery (for example more than four procedures a year), distal spread, or rapid regrowth
Dose	Intralesional injection of 5 to 7.5 mg/ml at the base of papillomas after debulking, at intervals of weeks, with a maximum per session related to body weight (often around 3 mg/kg)
Avoid	Renal impairment; pregnancy

2 CHAPTER TWO Using it in ENT

Indications.

2.1 Adjuvant therapy in papillomatosis

The mainstay is surgical debulking with microdebrider or laser while preserving normal tissue. Adjuvants are considered for severe disease: intralesional cidofovir, systemic **bevacizumab**, and the **HPV vaccine** (Gardasil), which may lengthen intervals between surgeries. Evidence from trials of cidofovir is mixed.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Systemically **nephrotoxic** (proximal tubule) and neutropenic; with intralesional use, systemic toxicity is uncommon but renal function and blood counts are checked. Concern that it may promote **dysplasia** has led to cautious use, with biopsies before each treatment.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The adjuvant options for **recurrent respiratory papillomatosis** and when to use them.
- The safety concerns with cidofovir.

4.2 Common pitfalls

COMMON PITFALLS

- Using cidofovir without biopsies to exclude dysplasia.

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